

Q1. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

What is the remainder when 853853853..... up to 100 digits is divided by 1001?

Q2. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

There are two pipes P_1 and P_2 , through which water flows into a tank at speeds of 2 m/s and 6 m/s respectively. If the cross-sectional areas of the pipes are 15 cm^2 and 25 cm^2 respectively, and it takes 40 minutes to fill the tank, then find the capacity of the tank (in kilolitres).

Q3. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

Two boys start simultaneously at the same point on a circular track and run along the track in the same direction. The point on the track at which they meet for the 5th time is the same as that at which they meet for the 17th time. If the ratio of the speed of the faster boy to that of the slower is $n : 1$, where n is a natural number, which of the following is not a possible value of n ?

☐ a) 2

☐ b) 6

☐ c) 5

☐ d) 4

Q4. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

If $f(x)$ is a quadratic polynomial, such that $f(5) = 35$ and $f(-5) = 15$, and $f(p) = f(q) = 0$, find $p + q$.

- ☐ a) 2
- ☐ b) -3
- ☐ c) 5
- ☐ d) Cannot be determined

Q5. DIRECTIONS *for questions 5 and 6:* Type in your answer in the input box provided below the question.

A circle and a regular hexagon are drawn in a plane. Find the maximum possible number of points of their intersection.

Q6. DIRECTIONS *for questions 5 and 6:* Type in your answer in the input box provided below the question.

Manishankar borrowed ₹5460 from a money lender and repaid it in three equal annual instalments, at a compound interest of 20% per annum, with each instalment being paid at the end of every year. What is the value (in ₹) of each instalment?

DIRECTIONS for questions 7 and 8: These questions are based on the following information.

For a real number x , let

$$f(x) = \frac{1}{1+x}, \text{ if } x \text{ is non-negative}$$

$$= 1+x, \text{ if } x \text{ is negative and}$$

$$f^n(x) = f(f^{n-1}(x)), \text{ for } n = 2, 3, \dots$$

Q7. DIRECTIONS for questions 7 and 8: Select the correct alternative from the given choices.

The value of $f(3) f^2(3) f^3(3) f^4(3) f^5(3)$ is

☐ a) $\frac{1}{8}$.

☐ b) 3.

☐ c) $\frac{1}{23}$.

☐ d) $\frac{1}{13}$.

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Q8. DIRECTIONS for questions 7 and 8: Select the correct alternative from the given choices.

For any integer $k \geq 2$, the value of $f^{k-1}(-k) + f^k(-k) + f^{k+1}(-k)$ is

- ☐ a) 1.
- ☐ b) 0.
- ☐ c) 0.2.
- ☐ d) 0.1.

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

Find the sum of all the terms common to the two sequences; 17, 21, 25, 417 and 16, 21, 26,, 466.

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

The question given below is followed by two statements I and II. Study the information given in the two statements and assess whether the information provided in them is sufficient to answer the question. Then select the correct answer option from among the choices given.

In a B-school, 30% of the students in a batch were male. If 10% of the female students have work experience, what percent of males have work experience?

- I. 20% of all the students in the batch have work experience.

- II. The number of males having work experience is 25% more than the number of females having work experience.

- ☐ a) if the question can be answered using either statement alone.
- ☐ b) if the question can be answered using both the statements together but not by using either statement alone.
- ☐ c) if the question cannot be answered even by using both the statements given.
- ☐ d) if the question can be answered using any one of the statements alone but not the other.

Q11. DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

The number of positive integral solutions to the equation $4x + 5y + 2z = 111$ is

Q12. DIRECTIONS for questions 12 to 18: Select the correct alternative from the given choices.

If x, y, z are real numbers such that $xyz = 1$, the expression $\frac{1}{1+x+y^{-1}} + \frac{1}{1+y+z^{-1}} + \frac{1}{1+z+x^{-1}}$ equates to

- ☐ a) $x + y + z$
- ☐ b) 1
- ☐ c) $x^{-1} + y^{-1} + z^{-1}$
- ☐ d) $\frac{3}{x + y + z}$

Q13. DIRECTIONS for questions 12 to 18: Select the correct alternative from the given choices.

Three positive integers, p , q and r , are such that $q > p > r$.

The following functions are now defined:

$$f(p, q, r) = \text{minimum} \{(p + r), (q + r)\}.$$

$$g(p, q, r) = \text{maximum} \{(q - p), (p - r)\}.$$

$$h(p, q, r) = \text{Average of } \{f(p, q, r), g(p, q, r)\}.$$

Which of the following is always false?

- ☐ a) $f(p, q, r) < g(p, q, r)$
- ☐ b) $h(p, q, r) < f(p, q, r)$
- ☐ c) $h(p, q, r) < g(p, q, r)$
- ☐ d) None of the above

Q14. DIRECTIONS *for questions 12 to 18:* Select the correct alternative from the given choices.
Consider the following definition of $f(x)$:

$$f(x) = f(f(x + 4)), \text{ if } x < 100 \text{ and}$$

$$f(x) = x - 2, \text{ if } x \geq 100.$$

Find the value of $f(50)$.

- ☐ a) 98
- ☐ b) 96
- ☐ c) 97
- ☐ d) 95

Q15. DIRECTIONS for questions 12 to 18: Select the correct alternative from the given choices.

Three lines represented by $x - y + 1 = 0$; $3x - 4y + 5 = 0$; and $3x + 2y - 7 = 0$ are drawn on the cartesian plane. The three lines

- ☐ a) form a triangle.
- ☐ b) are parallel.
- ☐ c) are concurrent.
- ☐ d) None of the above

Q16. DIRECTIONS *for questions 12 to 18:* Select the correct alternative from the given choices.

Two statements are considered to be inconsistent with each other, if the data given in one of the statements contradicts the data given in the other statement or if it is possible to arrive at conclusions that contradict each other using the two statements.

In the following question, there is a main statement (MS) followed by two additional statements, I and II. Study whether the MS is inconsistent or not inconsistent with each of the two statements. Then select the correct answer option from among the choices given.

MS:For the natural numbers N and D , the remainder when $2N$ is divided by D is 10.

I. The remainder when N is divided by $2D$ is 26.

II. The remainder when N is divided by D is 17.

- ☐ a) MS is inconsistent with I but is not inconsistent with II.
- ☐ b) MS is inconsistent with II but is not inconsistent with I.
- ☐ c) MS is inconsistent with I and also inconsistent with II.
- ☐ d) MS is inconsistent with neither I nor II.

Q17. DIRECTIONS *for questions 12 to 18:* Select the correct alternative from the given choices.

Three beakers A, B and C contain milk solutions of different concentrations C_1 , C_2 , and C_3 respectively, where $C_2 < C_1 < C_3$. Each beaker contains two litres of milk solution. One litre of solution from A is now taken out and poured in B. Then, one litre of the new solution from B is taken and poured in C. Finally, if the concentration in exactly two of the three beakers is the same, which of the following statements must necessarily be true?

☐ a) $4C_1 = 3C_2 + C_3$

☐ b) $2C_2 = C_3 + C_1$

☐ c) $4C_1 = C_2 + 3C_3$

☐ d) $3C_3 = C_1 + 2C_2$

Q18. DIRECTIONS *for questions 12 to 18:* Select the correct alternative from the given choices.

How many natural numbers less than 1000 can be expressed as the difference of two perfect squares in at least one way?

☐ a) 750

☐ b) 749

☐ c) 810

☐ d) 811

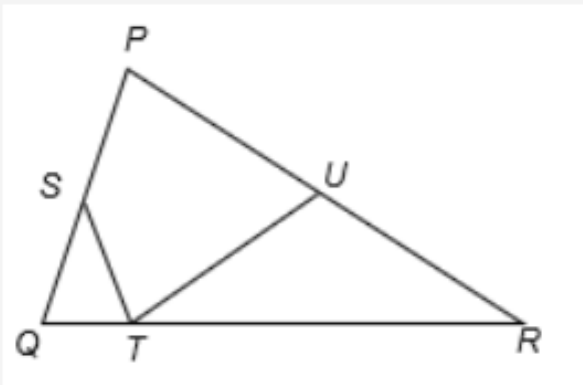
Q19. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

If $x > 1$, find the minimum value of $17\log_{30}x - 3\log_x 5 + 20\log_x 10 - 3\log_x 6 + 20\log_x 3$.

Q20. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

If $E = |x| + |x - 1| + |x - 2| + |x - 3| + |x - 4|$, for how many integral values of x is E less than 54?

Q21. DIRECTIONS for questions 21 and 22: Select the correct alternative from the given choices.



In the figure above, $\angle PQR = \angle PRQ + 30^\circ$ and $\angle QPR + \angle STU = 220^\circ$, $SQ = ST$ and $TU = UR$. Find $\angle PRQ$.

- ☐ a) 10°
- ☐ b) 15°
- ☐ c) 30°
- ☐ d) 20°

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

The set P is defined as the set of the first ten odd natural numbers. How many distinct pairs of numbers can be selected from the set P such that their sum is not less than 20?

☐ a) 25

☐ b) 20

☐ c) 15

☐ d) 45